IMPACT: Pathway-Aware and Equity-Audited Allocation under Scarce Resources

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Many public institutions must allocate scarce interventions under uncertainty: deciding not only who is at risk, but who is likely to benefit and whether the resulting allocation advances equity. In education, limited financial aid, advising, and academic support are often targeted before long-term outcomes are observed. Existing risk models prioritize students predicted to struggle, while direct treatment-effect models estimate responsiveness on the final outcome. However, these approaches often obscure the intermediate pathways through which interventions affect opportunity and provide limited visibility into who receives support and who benefits. We propose IMPACT, a pathway-aware and equity-audited framework for treatment allocation under scarce resources. IMPACT first identifies at-risk individuals using baseline covariates, estimates heterogeneous treatment effects on intermediate outcomes, and then evaluates whether predicted mediator pathways are likely to translate into final success. In our education application, we introduce Prospective IMPACT, a baseline-matched allocation rule that reconstructs proxy treated and untreated mediator profiles from similar historical students using only pre-treatment covariates. The candidate student's own first-year GPA and earned credits are not used at scoring time, making the framework compatible with prospective decision-making. To support accountable allocation, IMPACT reports both individual-level consistency and group-level equity diagnostics, including within-group selection rates, benefit shares, race-specific predicted allocation benefit, and fairness-adjusted utility. We evaluate IMPACT on a synthetic benchmark with known counterfactuals and on the Education Longitudinal Study of 2002, where financial aid is the intervention and first-year GPA and earned credits are mediators. On the ELS evaluation holdout, at a 10% budget, Prospective IMPACT identifies 13 at-risk students whose predicted success probability crosses the decision threshold, compared with 4 under direct final-outcome CATE and 0 under risk-only allocation. At a 30% budget, it identifies 16 threshold-crossing students, compared with 12 under direct CATE and 0 under risk-only allocation. The gains persist across threshold, matching-neighbor, score-weight, and fairness-penalty sensitivity analyses. These results show how pathway-aware allocation can make scarce-resource decisions more transparent, auditable, and equity-conscious.

Additional Key Words and Phrases: Heterogeneous Treatment Effects, Mediator Analysis, Education Policy, Index-based Treatment Allocation, Staged Intervention Modeling, Fair Resource Allocation

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1 Introduction

Public institutions routinely make allocation decisions under scarcity. In education, financial aid, advising, tutoring, and academic support cannot always be provided to every student who might benefit. Institutions must therefore decide who should receive limited support, when the decision must be made, and what evidence justifies the allocation. These decisions are not only predictive problems. They are equity-relevant policy choices that shape access to opportunity, persistence, and long-term educational attainment. This distinction is central to algorithmic decision-making in education. Standard risk models identify students who are predicted to struggle, but vulnerability is not the same as responsiveness to a specific intervention. A student may be highly at risk and yet unlikely to experience meaningful improvement from the available support. Conversely, another student may not be the highest-risk individual, but the intervention may improve an intermediate academic outcome, such as first-year GPA or earned credits, enough to change the student's predicted likelihood of success. Prior work has shown that financial resources and academic support can influence college access, persistence, and performance [3, 9, 14, 18, 43]. Yet many allocation systems still rely on either risk scores or direct treatment-effect estimates without explicitly modeling the pathways through which interventions change outcomes or auditing who receives and benefits from support.

This paper studies pathway-aware treatment allocation under scarce resources. We focus on settings in which an intervention affects intermediate outcomes, and those mediators subsequently influence a final outcome. In the education application considered here, receiving financial aid may affect first-year GPA and earned credits, which are in turn predictive of later student success. This mediated structure creates challenges for existing allocation strategies. A risk-only rule selects students with the lowest predicted probability of success, but high risk does not imply high benefit from treatment. A direct final-outcome CATE rule ranks students by estimated treatment effects on the final outcome, but it does not explain which intermediate academic pathways make the intervention effective. A mediator-only rule is more interpretable because it ranks students by predicted gains in GPA or credits, but large mediator gains are not necessarily outcome-relevant. A gain in earned credits matters most when it changes the predicted final outcome in a decision-relevant way. We propose IMPACT, a pathway-aware and equity-audited framework for allocating interventions under scarce resources. IMPACT first identifies an at-risk population using baseline covariates. It then estimates heterogeneous treatment effects on intermediate outcomes and evaluates whether the predicted mediator pathways are likely to translate into final success under a budget constraint. The key idea is that allocation should not be based only on who is most vulnerable, who has the largest final-outcome treatment effect, or who has the largest predicted mediator response. Instead, IMPACT asks whose treatment-induced pathway is predicted to be decisive for crossing a policy-relevant success threshold.

A central design requirement is prospective decision-making. In many real allocation settings, institutions must make decisions before post-treatment mediators are observed. In our education application, an institution allocating financial aid cannot use the student's own first-year GPA or earned credits at the time of scoring. We therefore introduce Prospective IMPACT, a baseline-matched allocation rule that imputes treated and untreated mediator potentials from similar historical students using only pre-treatment covariates. These imputed mediator potentials are then passed through an outcome model to estimate whether treatment is predicted to move the student across the success threshold. This design preserves the temporal structure of the decision problem and avoids relying on information that would not be available at the time of allocation. The framework is designed for accountable decision support rather than automated replacement of institutional judgment. It makes the allocation problem explicit by requiring a budget and comparing alternative ranking rules on the same at-risk population. It also treats equity as an object of measurement and deliberation rather than as an assumed property of predictive performance. IMPACT includes an individual-level consistency score, which discourages unstable treatment-effect estimates among similar individuals, and group-level equity diagnostics, including within-group selection rates, benefit shares, improvement rates, race-specific predicted allocation gain, and fairness-adjusted utility. These diagnostics do not prove that an allocation is fair. Rather, they make efficiency-equity tradeoffs visible, auditable, and comparable across allocation rules. This paper makes four contributions:

(1) We formulate mediated treatment allocation as a scarce-resource decision problem that combines risk identification, mediator-level treatment-effect estimation, prospective mediator-potential imputation, outcome-aware ranking, and budget-constrained allocation.

(2) We introduce Prospective IMPACT, an allocation rule that uses baseline-matched historical treated and untreated students to impute mediator potentials before the candidate student's own post-treatment mediators are observed.

(3) We develop an equity-auditing layer for pathway-aware allocation. The framework reports individual-level consistency and group-level diagnostics, including within-group selection rates, benefit shares, improvement rates,

race-specific predicted allocation gain, and fairness-adjusted utility. We also include an appendix extension that embeds the IMPACT score in a fairness-constrained allocation rule.

(4) We evaluate IMPACT on both a hard synthetic benchmark with known counterfactuals and the Education Longitudinal Study of 2002. The empirical results show that Prospective IMPACT is most valuable under scarce budgets and remains stable under threshold, matching-neighbor, composite-score weight, and fairness-penalty sensitivity analyses.

2 Motivating Example

Consider two students who are both identified as at risk of not completing a degree. Student A has the lower predicted probability of success and would be prioritized by a risk-only rule. However, similar historical students show only small changes in first-year GPA and earned credits after receiving financial aid, so the predicted treatment pathway is unlikely to move Student A above the success threshold. Student B has a slightly higher baseline success probability, but similar treated students show larger gains in earned credits relative to similar untreated students. When these reconstructed mediator profiles are passed through the outcome model, Student B's predicted success probability crosses the threshold. A risk-only rule selects Student A; a mediator-only rule may select either student; Prospective IMPACT selects Student B because the baseline-matched treatment pathway is predicted to be decisive. This example illustrates why vulnerability, responsiveness, pathway relevance, and equity auditing must be evaluated separately in scarce-resource allocation.

3 Related Work

Heterogeneous treatment effects. Conditional average treatment effect estimation has become central to personalized decision-making. Meta-learners [30], generalized random forests [1], orthogonal random forests [34], Bayesian non-parametric approaches [21], and double machine learning [10, 11] provide flexible tools for estimating heterogeneous effects in high-dimensional settings. Related methodological work emphasizes that heterogeneous effects require careful inference, validation, and comparison across learners [23, 24, 28]. These methods are powerful, but a CATE estimate alone does not solve the allocation problem. When resources are scarce, the relevant object is not only the treatment effect, but the ranking induced by that effect under a budget constraint.

Policy learning and constrained allocation. Empirical welfare maximization and policy learning study how to choose treatment rules from observational or experimental data [2, 26]. Recent work has emphasized ranking-based treatment allocation under budget constraints [25, 40]. IMPACT shares this allocative objective, but differs in two ways. First, it models mediator pathways rather than relying only on final-outcome responsiveness. Second, it reports fairness diagnostics at the point of allocation, making it possible to compare the utility and equity profile of different ranking rules. The allocation perspective is closely related to broader resource-prioritization settings such as triage, healthcare, social welfare programs, and public-service delivery under operational constraints [19, 27, 32, 33, 35, 37, 41, 42].

Mediation and causal pathways. Mediators are intermediate variables affected by the treatment and predictive of the final outcome. Causal mediation and potential-outcomes frameworks clarify the assumptions needed to interpret counterfactual pathways [4, 36, 39]. Prior work has emphasized that treatment heterogeneity can arise through distinct mechanisms [16, 17], and related causal-ML applications span health, marketing, and education [7, 13, 22, 38]. IMPACT operationalizes this idea for allocation by estimating treatment effects on mediators and then evaluating whether mediator pathways are predicted to affect the final outcome.

Fairness, interpretability, and educational decision-making. Fairness in resource allocation requires more than maximizing total welfare. In educational and public-sector settings, allocation rules can reproduce structural inequities if model outputs are not audited across groups [20, 31]. Educational decision-making also depends on definitions of success and institutional priorities, including performance metrics and resource allocation processes [15, 45]. We use individual-level consistency to discourage unstable treatment-effect estimates among similar individuals and add group-level diagnostics to evaluate allocation and benefit disparities. This connects causal allocation to the broader literature on equity-aware decision-making, interpretable machine learning, and resource prioritization [5, 6, 12, 29, 35, 44].

4 Problem Setup and Assumptions

Let $X_i \in \mathbb{R}^p$ denote pre-treatment baseline covariates for individual i, $T_i \in \{0, 1\}$ a binary treatment, $M_i \in \mathbb{R}^m$ a vector of post-treatment mediators, and $Y_i \in \{0, 1\}$ a final success outcome. In the education application (ELS), T_i indicates whether a student received a financial aid grant, the mediators are first-year GPA and earned credits, and Y_i is a binary indicator of degree completion or retention by the final ELS follow-up. The goal is to allocate a scarce intervention among students who are already identified as potentially vulnerable, while making both predicted benefit and equity consequences explicit. We first estimate a baseline success probability using a risk model $f_0(X_i)$ and define the at-risk set as $\mathcal{I}_{risk}(\theta) = \{i : \widehat{p}_{i,0} \le \theta\}, \quad \widehat{p}_{i,0} = f_0(X_i)$, where θ is a policy threshold. The threshold is not treated as a universal definition of need. Rather, it specifies the population for whom the institution wishes to consider additional support.

For each mediator j, IMPACT estimates a heterogeneous mediator effect, $\widehat{\tau}_j(X_i) \approx \mathbb{E}\left[M_{ij}(1) - M_{ij}(0) \mid X_i\right]$, where $M_{ij}(1)$ and $M_{ij}(0)$ denote the potential mediator values under treatment and no treatment, respectively. These mediator effects are not used alone to determine allocation. Instead, IMPACT evaluates whether the predicted mediator changes are likely to translate into a meaningful change in the final outcome.

Decision timing and information sets. The financial aid allocation problem is interpreted prospectively. At allocation time, the decision-maker observes baseline covariates for each student being considered for allocation, but not that student's first-year GPA, earned credits, or final outcome. Financial aid receipt is the treatment decision, while first-year GPA and earned credits are post-treatment mediators. Therefore, the main ELS allocation rule does not use realized mediator values from the candidate student. Instead, Prospective IMPACT uses only baseline covariates to identify similar historical treated and untreated students and reconstruct proxy treated and untreated mediator profiles from their observed post-treatment outcomes.

Causal assumptions. The framework relies on standard potential-outcomes assumptions. We assume consistency, no interference, treatment positivity, and conditional exchangeability of treatment assignment given observed baseline covariates. For pathway interpretation, we also assume that mediator-outcome relationships are not confounded by unobserved factors after conditioning on observed covariates and treatment. In the prospective matching step, we further require that comparable treated and untreated historical students exist within the relevant region of baseline covariate space. These assumptions are substantive and cannot be guaranteed by the model. They are especially strong in observational education data, where financial aid receipt may depend on unobserved factors such as application behavior, institutional discretion, family support, or unmet financial need. We therefore interpret the empirical results as estimated policy value under observed-covariate adjustment, not as experimental causal proof. To assess whether the analysis is credible on the observed support, we report predictive diagnostics, propensity overlap summaries, covariate-balance checks, and sensitivity analyses. In the ELS experiment, the estimated common-support interval contains 99.7% of observations.

Policy interpretation. The threshold θ plays two roles. First, it defines the at-risk population considered for allocation. Second, it defines a successful counterfactual crossing: a student is counted as improved when the predicted no-treatment outcome is below the threshold and the predicted treatment outcome is above it. This is a transparent policy definition rather than a claim that $\theta = 0.70$ is uniquely correct. The sensitivity analysis varies θ to test whether the conclusions depend on this choice.

5 The IMPACT Framework

IMPACT is a pathway-aware decision-support framework for allocating scarce interventions among individuals who have already been identified as potentially vulnerable. The framework separates three questions that are often conflated in algorithmic allocation: who is at risk, who is predicted to respond to an intervention, and whether the predicted response is likely to change the final outcome in a policy-relevant way. It also evaluates each allocation rule through both utility and equity diagnostics.

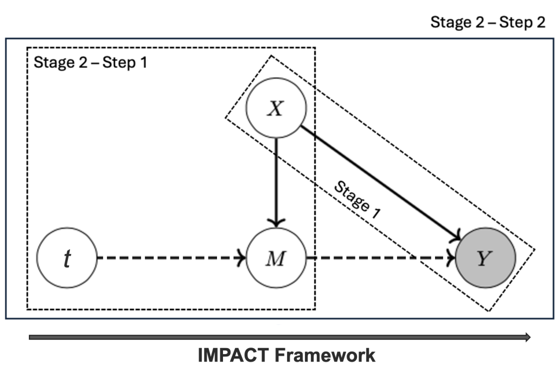
Figure 1 gives the system view and the fairness-efficiency evaluation. Stage 1 identifies the at-risk population from baseline covariates. Stage 2, Step 1 estimates treatment effects on post-treatment mediators. Stage 2, Step 2 evaluates whether reconstructed mediator pathways are predicted to translate into final success. The framework compares multiple allocation rankings under the same budget and reports utility and fairness diagnostics for each selected set.

Algorithm 1 summarizes the complete framework. The subsections below describe risk identification, mediator-aware ranking, prospective outcome-aware ranking, and fairness diagnostics.

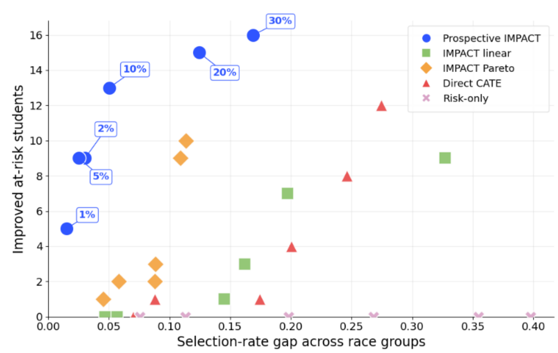
5.1 Risk Identification

The first stage identifies the at-risk population, Algorithm 2. This stage is intentionally separated from responsiveness estimation because vulnerability and treatment benefit are distinct. A student may be at high risk of not completing a degree without being the student most likely to benefit from a particular scarce intervention.

The threshold θ specifies the population eligible for allocation analysis. It should be interpreted as a policy parameter, not as a universal definition of need. All subsequent allocation rules are evaluated on the same at-risk set so that differences across methods reflect prioritization logic rather than changes in eligibility.



(a) IMPACT framework. Treatment T affects mediators M, baseline covariates X affect both mediators and final outcome Y, and Stage 2 evaluates how mediator pathways translate into success.



(b) Low-budget fairness-efficiency frontier. Each symbol represents an allocation rule at one low-budget level. Points closer to the upper-left combine higher benefit with lower racial disparity.

Fig. 1. Framework and fairness-efficiency evaluation. Panel (a) shows the pathway from treatment and baseline covariates to mediators and final outcomes. Panel (b) evaluates the low-budget tradeoff between predicted improvement among at-risk students and racial selection-rate gaps.

Algorithm 1 Overall IMPACT framework

Require: Baseline data $\mathcal{D}_0$, post-treatment historical data $\mathcal{D}_1$, candidate cohort C, threshold θ, budgets $\mathcal{B}$
Ensure: At-risk set, allocation rankings, selected sets, utility diagnostics, and fairness diagnostics

1: Fit baseline risk model $f_0(X)$ and define $\mathcal{I}_{risk} = \{i \in C : f_0(X_i) \leq \theta\}$
2: Estimate mediator effects $\{\widehat{\tau}_j(X)\}_{j=1}^m$ on $\mathcal{D}_1$
3: Compute local consistency $C_q(X_i)$ for each $i \in \mathcal{I}_{risk}$
4: Build mediator-aware rankings R^{lin} and R^{par} from $(\widehat{\tau}_1, \ldots, \widehat{\tau}_m, -C_q)$
5: Fit outcome model $f_1(X, M, T)$ on $\mathcal{D}_1$
6: **for** each $i \in \mathcal{I}_{risk}$ **do**
7: Reconstruct proxy treated and untreated mediator profiles $\widehat{M}_i(1)$ and $\widehat{M}_i(0)$ using baseline-matched historical treated and untreated students
8: Compute $\widehat{p}_i(1) = f_1(X_i, \widehat{M}_i(1), 1)$ and $\widehat{p}_i(0) = f_1(X_i, \widehat{M}_i(0), 0)$
9: Set $G_i = \widehat{p}_i(1) - \widehat{p}_i(0)$ and $I_i^{imp} = \mathbf{1}\{\widehat{p}_i(0) \leq \theta < \widehat{p}_i(1)\}$
10: **end for**
11: Build Prospective IMPACT ranking R^{pro} using predicted pathway gain, mediator response, and local consistency
12: Build benchmark rankings R^{cate} and R^{risk}
13: **for** each budget $b \in \mathcal{B}$ and ranking $r \in \{pro, lin, par, cate, risk\}$ **do**
14: Select the top $\lceil b|\mathcal{I}_{risk}|\rceil$ individuals under ranking r
15: Report improved counts, predicted gains, selection gaps, benefit shares, and fairness-adjusted utility
16: **end for**
17: **return** rankings, selected sets, utility diagnostics, and fairness diagnostics

Algorithm 2 Risk identification

Require: Historical baseline data $\mathcal{D}_0 = \{(X_i, Y_i)\}_{i=1}^n$, candidate cohort $\{X_i\}_{i=1}^N$, threshold θ
Ensure: At-risk set $\mathcal{I}_{risk}$

1: Fit risk model $f_0(X) \approx \Pr(Y = 1 \mid X)$ on $\mathcal{D}_0$
2: **for** each individual i in the candidate cohort **do**
3: $\widehat{p}_{i,0} \leftarrow f_0(X_i)$
4: **end for**
5: $\mathcal{I}_{risk} \leftarrow \{i : \widehat{p}_{i,0} \leq \theta\}$
6: **return** $\mathcal{I}_{risk}$

5.2 Mediator Effects and Individual Consistency

For each at-risk individual, IMPACT estimates treatment effects on post-treatment mediators. In the education application, the mediators are first-year GPA and earned credits. For mediator j, the estimated mediator effect is

$$\widehat{\tau}_j(X_i) \approx \mathbb{E}[M_{ij}(1) - M_{ij}(0) \mid X_i].$$

Mediator effects provide an interpretable description of how an intervention may operate, but they are not sufficient for allocation by themselves. A predicted gain in GPA or credits is most relevant when it changes the predicted final outcome.

IMPACT also computes a local consistency score to discourage unstable estimated effects among similar individuals. Let $\mathcal{N}_q(i)$ denote the q nearest neighbors of i in baseline covariate space. The implementation uses $q = 8$ for the consistency diagnostic. The consistency score is

$$C_q(X_i) = \frac{1}{m} \sum_{j=1}^{m} \frac{1}{q} \sum_{\ell \in \mathcal{N}_q(i)} \left| \widehat{\tau}_j(X_i) - \widehat{\tau}_j(X_\ell) \right|. \tag{1}$$

Lower values indicate that similar individuals have similar estimated mediator responses. This is an individual-level stability diagnostic related to the principle that similar individuals should not receive sharply different benefit estimates without evidence. It does not guarantee group parity, which is why IMPACT also reports group-level fairness metrics after allocation.

Algorithm 3 Mediator effect estimation and mediator-aware rankings

Require: At-risk set $\mathcal{I}_{risk}$, post-treatment historical data $\mathcal{D}_1 = \{(X_i, T_i, M_i, Y_i)\}$
Ensure: Linear and Pareto mediator-aware rankings

```
1: for each mediator j = 1, ..., m do
2:     Fit CATE model τ̂_j(X) ≈ E[M_j(1) − M_j(0) | X]
3: end for
4: for each i ∈ I_risk do
5:     Compute mediator effects τ̂_1(X_i), ..., τ̂_m(X_i)
6:     Compute local consistency C_q(X_i)
7:     S_i^med ← z(τ̂_GPA(X_i)) + z(τ̂_credits(X_i))
8:     Linear score S_i^lin ← S_i^med − z(C_q(X_i))
9: end for
10: Rank individuals by decreasing S_i^lin for Linear IMPACT
11: Rank individuals by Pareto fronts using objectives (τ̂_GPA, τ̂_credits, −C_q) for Pareto IMPACT
12: return Linear and Pareto rankings
```

5.3 Prospective Outcome-Aware Ranking

The outcome-aware component connects mediator pathways to the final outcome. In the prospective ELS allocation setting, the candidate student's own post-treatment mediators are not observed at the time of allocation. The main ELS rule therefore cannot use the candidate student's realized first-year GPA or earned credits.

Prospective IMPACT addresses this information constraint by reconstructing proxy treated and untreated mediator profiles from similar historical students. For each at-risk candidate student, we use baseline covariates to identify the k nearest historical treated students and the k nearest historical untreated students. The observed mediator outcomes of these matched historical students are averaged to construct proxy mediator profiles:

$$\widehat{M}_i(1) = \frac{1}{k} \sum_{\ell \in \mathcal{N}_k^1(i)} M_\ell, \qquad \widehat{M}_i(0) = \frac{1}{k} \sum_{\ell \in \mathcal{N}_k^0(i)} M_\ell, \tag{2}$$

where $\mathcal{N}_k^1(i)$ and $\mathcal{N}_k^0(i)$ are the k nearest historical treated and untreated students to i in baseline covariate space. The main ELS analysis uses $k = 5$ and reports sensitivity for $k = 1, \ldots, 5$.

The reconstructed profiles $\widehat{M}_i(1)$ and $\widehat{M}_i(0)$ should be interpreted as proxy treated and untreated mediator profiles, not observed counterfactual ground truth. They summarize what happened to similar historical students under treatment and no treatment, using only information that would be available at allocation time.

A post-treatment outcome model $f_1(X, M, T)$ then estimates

$$\widehat{p}_i(1) = f_1(X_i, \widehat{M}_i(1), 1), \quad \widehat{p}_i(0) = f_1(X_i, \widehat{M}_i(0), 0). \tag{3}$$

The predicted pathway gain is $G_i = \widehat{p}_i(1) - \widehat{p}_i(0)$, and an individual is counted as improved if $\widehat{p}_i(0) \le \theta < \widehat{p}_i(1)$. This improvement indicator captures whether the reconstructed treatment pathway is predicted to move the student across the policy-relevant success threshold.

The final prospective score is defined as

$$A_i(w_m, w_c) = z(G_i) + w_m z(S_i^{med}) - w_c z(C_q(X_i)), \qquad S_i^{med} = z(\hat{\tau}_{GPA}(X_i)) + z(\hat{\tau}_{credits}(X_i)).$$

The main specification sets $w_m = w_c = 0.25$, so predicted final-outcome gain remains the dominant component while mediator response and local consistency act as moderate stabilizers. These auxiliary weights are not tuned to maximize the main result; Appendix A.1 reports sensitivity over alternative values of w_m and w_c. Here, $z(\cdot)$ denotes standardization over the at-risk cohort being ranked.

Algorithm 4 Baseline-Matched Prospective IMPACT

Require: Historical data $\mathcal{D}_1 = \{(X_i, T_i, M_i, Y_i)\}_{i=1}^n$, at-risk set $\mathcal{I}_{risk}$, outcome model f_1, threshold θ, matches k
Ensure: Prospective IMPACT ranking and improvement indicator

```
Split historical data into treated set H^1 = {i : T_i = 1} and untreated set H^0 = {i : T_i = 0}
for each i ∈ I_risk do
    Find k nearest historical treated students N_k^1(i) ⊂ H^1 using baseline covariates only
    Find k nearest historical untreated students N_k^0(i) ⊂ H^0 using baseline covariates only
    M̂_i(1) ← k^{-1} Σ_{ℓ∈N_k^1(i)} M_ℓ
    M̂_i(0) ← k^{-1} Σ_{ℓ∈N_k^0(i)} M_ℓ
    p̂_i(1) ← f_1(X_i, M̂_i(1), 1)
    p̂_i(0) ← f_1(X_i, M̂_i(0), 0)
    G_i ← p̂_i(1) − p̂_i(0)
    I_i^imp ← 1{p̂_i(0) ≤ θ < p̂_i(1)}
    A_i ← z(G_i) + 0.25 z(S_i^med) − 0.25 z(C_q(X_i))
end for
Rank individuals by decreasing A_i
return Prospective IMPACT ranking and I_i^imp
```

5.4 Allocation Rules and Fairness Diagnostics

We compare five allocation rules. *Prospective IMPACT* ranks individuals by A_i, the standardized score combining predicted pathway gain, mediator response, and local consistency. *IMPACT Linear* and *IMPACT Pareto* are mediator-only ablations. They test whether ranking by predicted mediator response alone is sufficient, without evaluating whether those mediator pathways translate into final success. *Direct final-outcome CATE* ranks individuals by estimated treatment responsiveness on the final outcome. *Risk-only* selects individuals with the lowest predicted baseline success probability.

For a selected set $\mathcal{S}_b$ at budget b, we report the improved count $\sum_{i\in\mathcal{S}_b} I_i^{imp}$ and the improvement rate among selected individuals. Let Z_i denote the protected or audited group attribute, such as race, and let $\mathcal{I}_g = \{i \in \mathcal{I}_{risk} : Z_i = g\}$ denote the at-risk individuals in group g. For each group g, we compute

$$\text{SelectionRate}_g = \frac{|\mathcal{S}_b \cap \mathcal{I}_g|}{|\mathcal{I}_g|}, \quad \text{BenefitShare}_g = \frac{\sum_{i\in\mathcal{S}_b\cap\mathcal{I}_g} I_i^{imp}}{\sum_{i\in\mathcal{S}_b} I_i^{imp}}. \tag{4}$$

When the denominator of the benefit share is zero, the benefit-share diagnostic is reported as undefined rather than interpreted as evidence of parity.

The selection-rate gap is then defined as $\text{SelectionGap}(\mathcal{S}_b) = \max_g \text{SelectionRate}_g - \min_g \text{SelectionRate}_g$. We also report group-specific improvement rates and mean predicted allocation gains among selected individuals within each group. These diagnostics show whether predicted benefits are concentrated in particular groups or distributed more evenly across the selected population. Finally, we define fairness-adjusted utility as

$$U_\lambda(\mathcal{S}_b) = \sum_{i \in \mathcal{S}_b} I_i^{imp} - \lambda\, |\mathcal{S}_b|\, \mathrm{SelectionGap}(\mathcal{S}_b), \tag{5}$$

with $\lambda \in \{0.25, 0.50, 1.00\}$ in sensitivity analysis. This metric is not treated as a universal fairness measure. It is a compact way to examine how the ranking performs when a penalty is placed on racial allocation imbalance.

Protected attributes are used for auditing in the main specification, not for treatment assignment. The purpose of the fairness diagnostics is not to certify that an allocation is fair, but to make allocation and benefit disparities visible before deployment. The appendix reports an optional fairness-constrained allocation layer that maximizes the Prospective IMPACT score subject to an upper bound on within-race selection-rate gaps.

6 Experimental Design

6.1 Benchmarks

ELS 2002. We evaluate IMPACT on the Education Longitudinal Study of 2002, a nationally representative longitudinal study of U.S. students [8]. The treatment is financial aid grant receipt. The mediators are first-year GPA and earned credits. The final outcome is a binary degree-completion or retention indicator derived from the ELS highest-education-attainment measure at the final follow-up. Baseline covariates include demographic, academic, family, school, and expectation variables observed before or around college entry. The main ELS allocation evaluation uses a mixed internal evaluation holdout from the historical ELS file. The training sample contains 4,567 students, and the mixed policy holdout contains 1,523 students, including 893 treated and 630 untreated students. This design allows allocation policies to be compared on a common historical support. For Prospective IMPACT, the candidate student's realized first-year GPA and earned credits are not used in scoring. Instead, treated and untreated proxy mediator profiles are reconstructed from baseline-matched historical treated and untreated students. The ELS evaluation should be interpreted as an internal historical policy evaluation rather than as evidence from deployment on a newly arriving cohort. The evaluation holdout is separated from model fitting, but it is drawn from the same historical data source. This design is useful for comparing allocation rules under common support, but it does not eliminate concerns about temporal drift, institutional heterogeneity, unmeasured confounding, or changes in aid policy that may arise in prospective deployment.

Synthetic. We also evaluate IMPACT on a hard synthetic benchmark with 5,000 observations, eight baseline covariates, a binary group indicator, a binary treatment, two post-treatment mediators, and a binary final outcome. Treatment assignment depends on baseline covariates and group membership, creating measured confounding. Treatment effects are heterogeneous because the intervention shifts the two mediators differently across individuals, and the final outcome is generated from a nonlinear function of covariates, treatment, and mediators. The benchmark stores the true untreated and treated outcome probabilities, $p_i(0)$ and $p_i(1)$, so each rule can be evaluated using true policy value rather than only estimated counterfactual crossings. This experiment tests the pathway-aware allocation logic in a setting where the true allocation benefit is known, while the ELS application evaluates the same logic in a realistic education setting.

6.2 Models, Budgets, and Metrics

The ELS implementation uses gradient-boosted models for risk prediction, mediator response estimation, propensity estimation, and post-treatment outcome prediction. Mediator CATEs and the direct final-outcome CATE benchmark are implemented as T-learners fitted separately for treated and untreated students. Historical data are split into training and holdout portions for model diagnostics, and the selected models are then refit on the historical training sample before scoring the mixed policy holdout. The evaluation holdout is stratified by treatment, outcome, and race where feasible. Reconstructed GPA values are clipped to the feasible $[0, 4]$ range, and reconstructed credits are clipped to the observed support. All allocation rules are evaluated on the same at-risk set and under the same budgets. The primary utility metric is the number of improved at-risk individuals selected under a fixed budget, where improvement is defined as crossing the policy-relevant success threshold under the predicted treatment pathway. We focus on budgets from 1% to 30%, because allocation methods are most consequential when resources are scarce. The main text reports full-budget curves, threshold sensitivity, matching-neighbor sensitivity, composite-score weight sensitivity, fairness-penalty sensitivity, and race-level diagnostics. The implementation is modular: risk prediction, mediator-response estimation, baseline matching, outcome scoring, and fairness auditing are separable steps that can be replaced or extended in other domains.

7 Results

Table 1 summarizes the ELS design, predictive, and overlap diagnostics. The risk model obtains an AUC of 0.819, the post-treatment outcome model obtains an AUC of 0.859, and the estimated propensity model obtains an AUC of 0.757. The estimated common-support interval contains 99.7% of observations. These diagnostics do not establish causal identification, but they indicate that the comparison is not driven by unusable predictive models or severe lack of observed overlap.

The treated and untreated groups occupy largely overlapping regions of the estimated propensity-score distribution. Their distributions are not identical, which is expected in observational education data, but the common-support and tail-share diagnostics in Table 1 support comparison on the observed support. Additional balance diagnostics show that inverse-propensity weighting reduces the mean absolute standardized mean difference from 0.086 to 0.035 and raises the share of covariates below 0.10 absolute imbalance from 70.6% to 95.4%. Because the main improvement criterion depends on predicted probabilities crossing a threshold, calibration is also relevant. We therefore report Brier scores alongside AUC and interpret threshold-crossing counts as model-based estimates of policy value rather than observed causal outcomes. Throughout the results, the estimates should be read as conditional on observed covariates, overlap, and the modeling assumptions described above.

7.1 Allocation Performance under Scarce Budgets

Figure 2 and Table 2 report the main prospective ELS allocation results. At a 10% budget and $\theta = 0.70$, Prospective IMPACT identifies 13 at-risk students whose predicted success probability crosses the decision threshold, compared with 3 under IMPACT Linear, 3 under IMPACT Pareto, 4 under direct final-outcome CATE, and 0 under risk-only allocation. At a 30% budget, Prospective IMPACT identifies 16 threshold-crossing students, compared with 9 under IMPACT Linear, 10 under IMPACT Pareto, 12 under direct CATE, and 0 under risk-only allocation. These estimates are

Table 1. Design, predictive, and overlap diagnostics for the ELS. The policy holdout contains both treated and untreated students.

Metric	Value	Metric	Value
Historical training sample size	4,567	Outcome model AUC	0.859
Mixed-policy holdout sample size	1,523	Outcome model Brier	0.151
Holdout treated count	893	Propensity model AUC	0.757
Holdout untreated count	630	Propensity model Brier	0.196
At-risk student count	1,053	Share in common support	0.997
Risk model AUC	0.819	Share below 0.05 propensity	0.001
Risk model Brier	0.173	Share above 0.95 propensity	0.011

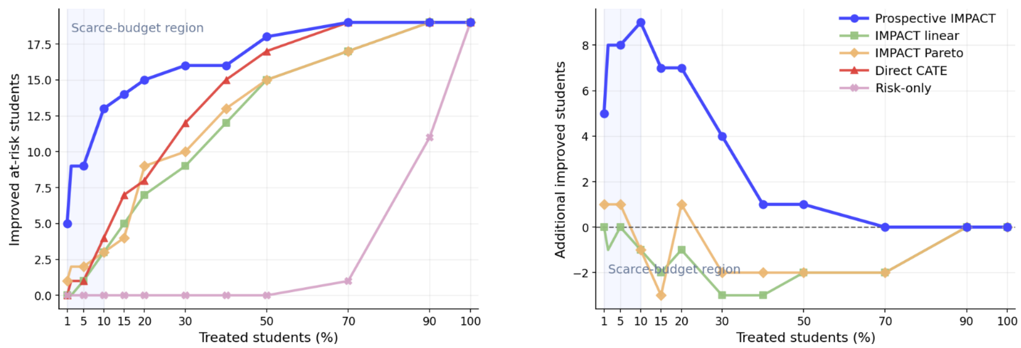


Fig. 2. ELS allocation performance across the full budget range. Panel (a) reports improved at-risk students selected by each policy. Panel (b) reports the additional improved students achieved by the IMPACT variants relative to the direct final-outcome CATE.

Table 2. ELS prospective allocation performance under scarce budgets. Entries report the number of at-risk students whose predicted success probability crosses the decision threshold under the allocated intervention. In parentheses improved students

Budget	Prospective IMPACT	IMPACT Linear	IMPACT Pareto	Direct CATE	Risk-only
1%	5 (45.5%)	0 (0.0%)	1 (9.1%)	0 (0.0%)	0 (0.0%)
2%	9 (40.9%)	0 (0.0%)	2 (9.1%)	1 (4.5%)	0 (0.0%)
5%	9 (17.0%)	1 (1.9%)	2 (3.8%)	1 (1.9%)	0 (0.0%)
10%	13 (12.3%)	3 (2.8%)	3 (2.8%)	4 (3.8%)	0 (0.0%)
20%	15 (7.1%)	7 (3.3%)	9 (4.3%)	8 (3.8%)	0 (0.0%)
30%	16 (5.1%)	9 (2.8%)	10 (3.2%)	12 (3.8%)	0 (0.0%)

prospective in the sense that the ranking uses only baseline covariates and reconstructed mediator profiles derived from historical matches, not the candidate student's realized post-treatment mediators.

The results show why the outcome-aware pathway step matters. Linear and Pareto IMPACT use mediator response information, but they do not directly evaluate whether predicted mediator changes alter the final success prediction. Direct final-outcome CATE targets responsiveness on the final outcome, but it does not expose the academic pathway through which the intervention is predicted to operate. Prospective IMPACT combines these two perspectives: it reconstructs treated and untreated mediator profiles from baseline-matched historical students and then evaluates whether those pathway differences are predicted to move the student across the success threshold. Panel (b) of Figure 2 shows this as an incremental allocation gain relative to direct CATE.

The comparison also illustrates the difference between vulnerability and predicted benefit. Risk-only allocation is need-focused and transparent, but it performs poorly as a benefit-targeting rule because the lowest predicted baseline success probability does not necessarily identify students most likely to cross the success threshold under treatment. Direct CATE improves on risk-only allocation by targeting final-outcome responsiveness, but it does not provide a pathway-level explanation. Prospective IMPACT retains a pathway interpretation while prioritizing students whose reconstructed treatment pathway is predicted to be outcome-relevant.

7.2 Fairness and Equity Diagnostics

Figure 1b visualizes the tradeoff between utility and allocation disparity, and Table 3 reports compact utility and fairness diagnostics. At a 10% budget, Prospective IMPACT has a selection-rate gap of 0.050, compared with 0.162 for

Table 3. ELS utility and fairness diagnostics for the prospective allocation rules.

Budget	Method	Improved	Selection-rate gap	Avg. gain
10%	Prospective IMPACT	13	0.050	0.279
10%	IMPACT Linear	3	0.162	0.150
10%	IMPACT Pareto	3	0.088	0.123
10%	Direct CATE	4	0.201	0.072
10%	Risk-only	0	0.268	0.015
30%	Prospective IMPACT	16	0.169	0.189
30%	IMPACT Linear	9	0.327	0.124
30%	IMPACT Pareto	10	0.113	0.114
30%	Direct CATE	12	0.274	0.061
30%	Risk-only	0	0.398	0.028

IMPACT Linear, 0.088 for IMPACT Pareto, 0.201 for direct CATE, and 0.268 for risk-only allocation. At a 30% budget, the selection-rate gap is 0.169 for Prospective IMPACT, compared with 0.327 for IMPACT Linear, 0.113 for IMPACT Pareto, 0.274 for direct CATE, and 0.398 for risk-only allocation. These results should be interpreted carefully. Prospective IMPACT is not a fairness guarantee, and the score is not designed to enforce group parity. Rather, the favorable equity diagnostics are empirical properties of this application. The main contribution is that the framework makes allocation disparities visible alongside predicted utility. This is important for equity-relevant decision-making: a rule that improves aggregate predicted outcomes may still systematically over-select, under-select, or under-benefit particular racial groups. IMPACT is designed to surface these patterns before deployment.

Figure 3 adds a distributional view to the aggregate fairness metrics. The selected benefits are shaped by the underlying composition of the at-risk set, but the comparison makes the allocation transparent. Readers can inspect how many improved students come from each racial group and how the pattern changes under Prospective, Linear, and Pareto IMPACT rankings. Thus, IMPACT reports both aggregate utility and race-specific allocation and benefit diagnostics rather than relying on a single aggregate performance measure.

7.3 Synthetic Ground-Truth Validation

The hard synthetic experiment validates the pathway-aware allocation logic when true counterfactuals are known. Table 4 reports the low-budget validation results. The outcome-aware synthetic rule achieves the largest true allocation benefit from the 2% budget onward. At the 1% budget, Pareto is marginally higher, with a true allocation benefit of

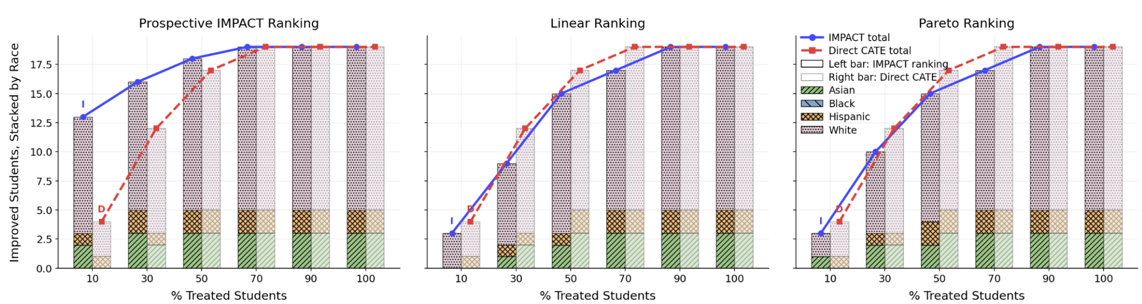


Fig. 3. Race-stacked improved students for the IMPACT rankings.

4.42 compared with 4.40 for the outcome-aware rule. This exception is useful because it shows that the comparison is not hard-coded to favor the outcome-aware method. The broader pattern is that outcome-aware pathway simulation selects individuals with higher ground-truth policy value when budgets are scarce.

7.4 Sensitivity to Thresholds, Matching, Score Weights, and Fairness Penalties

The main ELS analysis uses $\theta = 0.70$, but this threshold is a policy parameter rather than a universal definition of success. Figure 4 varies θ from 0.40 to 0.70. The absolute number of improved students changes because the at-risk set

Table 4. Hard synthetic validation using known ground-truth policy value. Entries report true policy value under each allocation rule. The outcome-aware synthetic rule is best from the 2% budget onward, while Pareto is marginally higher at the 1% budget.

Budget	Outcome-aware	Linear	Pareto	Direct CATE	Risk-only
1%	4.40	3.86	4.42	3.52	1.10
2%	8.58	7.96	7.72	6.19	2.27
5%	20.16	18.40	17.94	16.00	6.38
10%	39.00	35.63	34.89	30.05	13.96
20%	72.80	66.27	67.44	59.92	33.80
30%	103.44	96.96	97.26	88.78	57.44

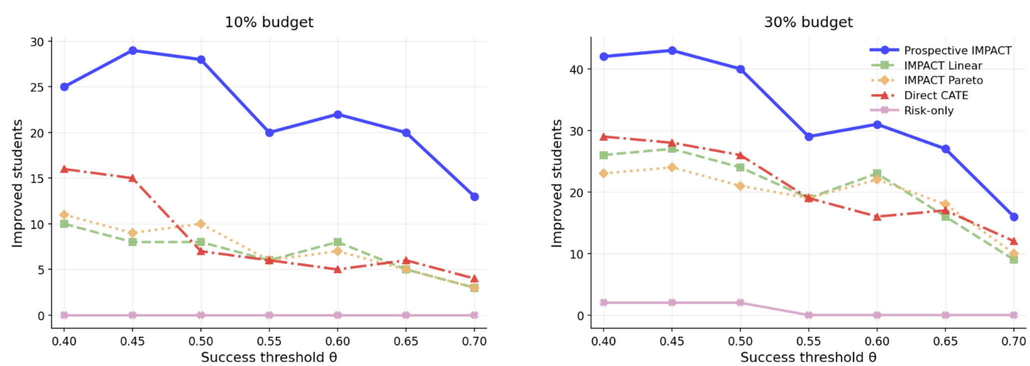


Fig. 4. Sensitivity to the success threshold θ. The figure varies the threshold used to define at-risk status and treatment-induced improvement. Prospective IMPACT remains robust across the tested thresholds at both budget levels.

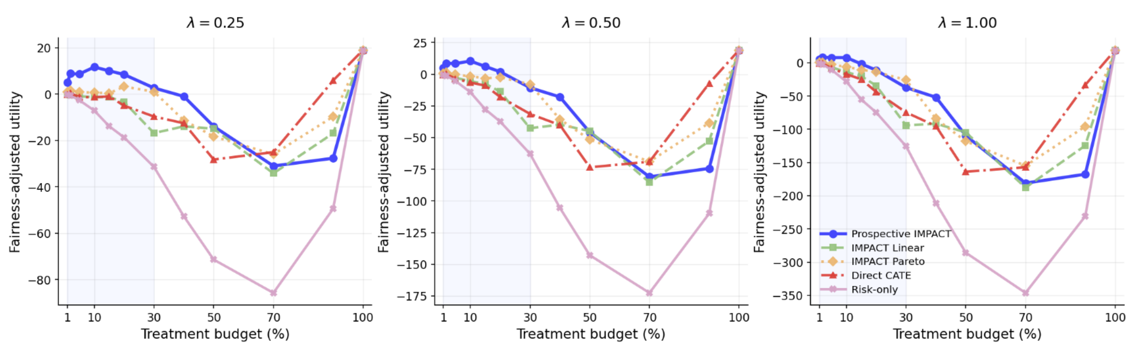


Fig. 5. Sensitivity to the fairness penalty λ. The figure varies the penalty placed on racial allocation imbalance in the fairness-adjusted utility. Prospective IMPACT remains robust under weaker, moderate, and stronger fairness penalties.

and the crossing definition change. However, the comparative pattern remains stable: Prospective IMPACT remains the strongest or among the strongest allocation rules across the tested thresholds and budget levels.

The matching-neighbor sensitivity further supports this conclusion. Across $k = 1, \ldots, 5$, thresholds from 0.40 to 0.70, and all budget levels, Prospective IMPACT beats the direct CATE benchmark in 89.0% of improved-count comparisons, ties in 11.0%, and loses in 0.0%. Against risk-only allocation, it wins in 91.7% of comparisons, ties in 8.3%, and loses in 0.0%. The mean improved-count advantage is 14.01 students relative to direct CATE and 31.91 students relative to risk-only allocation. We also test whether the composite score depends on the fixed auxiliary weights assigned to mediator response and local consistency. Across a grid of 25 mediator-response and consistency-weight specifications, Prospective IMPACT remains above the direct CATE benchmark in all tested specifications at both the 10% and 30% budgets. The canonical weights should therefore be interpreted as moderate stabilizing weights rather than tuned parameters. The appendix reports the full score-weight sensitivity table.

Table 6 and Figure 5 vary the fairness-penalty parameter λ in the fairness-adjusted utility. Prospective IMPACT remains favorable relative to the main baselines under mild, moderate, and stronger fairness penalties. This does not imply that any particular value of λ is ethically correct. Rather, it shows that the main ranking is not driven by a single arbitrary fairness-penalty choice.

8 Conclusion

IMPACT reframes mediated treatment-effect estimation as a scarce-resource allocation problem. The framework identifies at-risk individuals, estimates mediator-level treatment effects, imputes prospective mediator potentials, and ranks individuals by predicted final-outcome gain under a budget constraint. In the ELS financial-aid application, Prospective IMPACT uses baseline-matched historical treated and untreated students to impute mediator potentials before first-year outcomes are observed. This makes the allocation rule compatible with the information available at the time of aid targeting. The framework also adds explicit fairness diagnostics, showing that utility gains can be evaluated

Table 5. Matching-neighbor sensitivity for Prospective IMPACT across $k = 1, \ldots, 5$, thresholds from 0.40 to 0.70, and the full budget grid. Entries report win, tie, and loss rates in improved-count comparisons.

Baseline	Comparisons	Win	Tie	Loss	Mean improved advantage
Direct CATE	420	89.0%	11.0%	0.0%	14.01
Risk-only	420	91.7%	8.3%	0.0%	31.91

Table 6. Sensitivity of fairness-adjusted utility to the penalty parameter λ under the prospective ELS allocation design.

Budget	Method	$\lambda = 0.25$	$\lambda = 0.50$	$\lambda = 1.00$
10%	Prospective IMPACT	11.67	10.34	7.68
10%	IMPACT Linear	-1.29	-5.58	-14.16
10%	IMPACT Pareto	0.67	-1.67	-6.33
10%	Direct CATE	-1.32	-6.63	-17.26
10%	Risk-only	-7.10	-14.20	-28.39
30%	Prospective IMPACT	2.65	-10.70	-37.40
30%	IMPACT Linear	-16.83	-42.66	-94.32
30%	IMPACT Pareto	1.04	-7.93	-25.85
30%	Direct CATE	-9.68	-31.36	-74.72
30%	Risk-only	-31.41	-62.83	-125.66

alongside racial allocation and benefit disparities. The synthetic and sensitivity analyses support the robustness of the main findings. More broadly, IMPACT provides a practical and auditable template for causal decision support in settings where resources are scarce, interventions operate through intermediate pathways, and equity is central to evaluation.

References

[1] Susan Athey, Julie Tibshirani, and Stefan Wager. 2019. Generalized random forests. *The Annals of Statistics* 47, 2 (2019), 1148–1178. doi:10.1214/18-AOS1709

[2] Susan Athey and Stefan Wager. 2021. Policy learning with observational data. *Econometrica* 89, 1 (2021), 133–161.

[3] Laura S. Bakosh, Renee M. Snow, Jutta M. Tobias Mortlock, Janice L. Houlihan, and Celestina Barbosa-Leiker. 2016. Maximizing mindful learning: Mindful awareness intervention improves elementary school students' quarterly grades. *Mindfulness* 7, 1 (2016), 59–67. doi:10.1007/s12671-015-0387-6

[4] Debopam Bhattacharya and Pascaline Dupas. 2012. Inferring welfare maximizing treatment assignment under budget constraints. *Journal of Econometrics* 167, 1 (2012), 168–196. doi:10.1016/j.jeconom.2011.11.007

[5] Niclas Boehmer, Yash Nair, Sanket Shah, Lucas Janson, Aparna Taneja, and Milind Tambe. 2024. Evaluating the Effectiveness of Index-Based Treatment Allocation. arXiv:2402.11771 [cs.LG]

[6] Niclas Boehmer, Yash Nair, Sanket Shah, Lucas Janson, Aparna Taneja, and Milind Tambe. 2025. Evaluating Index-based Treatment Allocation in Underresourced Communities. In *Proceedings of the AAAI Conference on Artificial Intelligence*, Vol. 39. 27849–27857. doi:10.1609/aaai.v39i27.35001

[7] Hakan Şat Bozcuk and Mustafa Serkan Alemdar. 2024. Solving the puzzle of quality of life in cancer: integrating causal inference and machine learning for data-driven insights. *Health and Quality of Life Outcomes* 22, 1 (2024), 60. doi:10.1186/s12955-024-02274-7

[8] Robert Bozick, Eric Lauff, and John Wirt. 2007. *Education Longitudinal Study of 2002 (ELS: 2002): A First Look at the Initial Postsecondary Experiences of the High School Sophomore Class of 2002.* Technical Report. National Center for Education Statistics.

[9] Priyadarshini Chaplot, Darla M. Cooper, Rob Johnstone, and Kelley Karandjeff. 2015. *Beyond Financial Aid: How colleges can strengthen the financial stability of low-income students and improve student outcomes.* Technical Report.

[10] Victor Chernozhukov, Denis Chetverikov, Mert Demirer, Esther Duflo, Christian Hansen, Whitney Newey, and James Robins. 2017. Double/debiased machine learning for treatment and structural parameters. arXiv preprint arXiv:1608.00060v6.

[11] Victor Chernozhukov, Denis Chetverikov, Mert Demirer, Esther Duflo, Christian Hansen, Whitney Newey, and James M. Robins. 2018. Double/debiased machine learning for treatment and structural parameters. *The Econometrics Journal* 21, 1 (2018), C1–C68.

[12] Finale Doshi-Velez and Been Kim. 2017. Towards a Rigorous Science of Interpretable Machine Learning. arXiv preprint arXiv:1702.08608.

[13] Bao Duong, Manisha Senadeera, Toan Nguyen, Melanie Nichols, Kathryn Backholer, Steven Allender, and Thin Nguyen. 2024. Utilising causal inference methods to estimate effects and strategise interventions in observational health data. *PLOS ONE* 19, 12 (2024), e0314761.

[14] Susan Dynarski. 2008. Building the stock of college-educated labor. *Journal of Human Resources* 43, 3 (2008), 576–610.

[15] Peter T. Ewell. 1999. Linking performance measures to resource allocation: Exploring unmapped terrain. *Quality in Higher Education* 5, 3 (1999), 191–209.

[16] Paul J. Ferraro and Juan José Miranda. 2013. Heterogeneous treatment effects and causal mechanisms in non-pecuniary, information-based environmental policies: Evidence from a large-scale field experiment. *Resource and Energy Economics* 35, 3 (2013), 356–379.

[17] Jiawei Fu. 2024. Extract mechanisms from heterogeneous effects: Identification strategy for mediation analysis. arXiv preprint arXiv:2403.04131.

[18] Sara Goldrick-Rab, Robert Kelchen, Douglas N. Harris, and James Benson. 2016. Reducing income inequality in educational attainment: Experimental evidence on the impact of financial aid on college completion. *Amer. J. Sociology* 121, 6 (2016), 1762–1817.

[19] Katerina Gousia, Hansel Teo, Stacey Rand, and Florin Vadean. 2024. Determining the cost-effectiveness of home care: A scoping review. *Journal of Long-Term Care* (2024), 446–463. doi:10.31389/jltc.314

[20] Lalla Aïda Guindo, Monika Wagner, Rob Baltussen, Donna Rindress, Janine van Til, Paul Kind, and Mireille M. Goetghebeur. 2012. From efficacy to equity: Literature review of decision criteria for resource allocation and healthcare decisionmaking. *Cost Effectiveness and Resource Allocation* 10, 9 (2012). doi:10.1186/1478-7547-10-9

[21] Jennifer L. Hill. 2011. Bayesian nonparametric modeling for causal inference. *Journal of Computational and Graphical Statistics* 20, 1 (2011), 217–240.

[22] Günter J. Hitsch, Sanjog Misra, and Walter W. Zhang. 2024. Heterogeneous treatment effects and optimal targeting policy evaluation. *Quantitative Marketing and Economics* 22, 2 (2024), 115–168.

[23] Kosuke Imai and Michael Lingzhi Li. 2023. Experimental evaluation of individualized treatment rules. *J. Amer. Statist. Assoc.* 118, 541 (2023), 242–256.

[24] Kosuke Imai and Michael Lingzhi Li. 2025. Statistical Inference for Heterogeneous Treatment Effects Discovered by Generic Machine Learning in Randomized Experiments. *Journal of Business & Economic Statistics* 43, 1 (2025), 256–268. doi:10.1080/07350015.2024.2358909

[25] Fahad Kamran, Maggie Makar, and Jenna Wiens. 2024. Learning to Rank for Optimal Treatment Allocation Under Resource Constraints. In *Proceedings of the 27th International Conference on Artificial Intelligence and Statistics (Proceedings of Machine Learning Research, Vol. 238).* PMLR, 3727–3735. https://proceedings.mlr.press/v238/kamran24a.html

[26] Toru Kitagawa and Aleksey Tetenov. 2018. Who should be treated? Empirical welfare maximization methods for treatment choice. *Econometrica* 86, 2 (2018), 591–616.

[27] Eike-Henner W. Kluge. 2007. Resource allocation in healthcare: implications of models of medicine as a profession. *Medscape General Medicine* 9, 1 (2007), 57.

[28] Michael C. Knaus, Michael Lechner, and Anthony Strittmatter. 2021. Machine Learning Estimation of Heterogeneous Causal Effects: Empirical Monte Carlo Evidence. *The Econometrics Journal* 24, 1 (2021), 134–161. doi:10.1093/ectj/utaa014

[29] Richard L. Kravitz, Naihua Duan, and Joel Braslow. 2004. Evidence-based medicine, heterogeneity of treatment effects, and the trouble with averages. *The Milbank Quarterly* 82, 4 (2004), 661–687.
[30] Sören R. Künzel, Jasjeet S. Sekhon, Peter J. Bickel, and Bin Yu. 2019. Metalearners for estimating heterogeneous treatment effects using machine learning. *Proceedings of the National Academy of Sciences* 116, 10 (2019), 4156–4165.
[31] Tai Le Quy. 2024. *Fairness-aware Machine Learning in Educational Data Mining.* Ph. D. Dissertation. Leibniz University Hannover. doi:10.15488/15832
[32] Douglas E. Levy, Deepinder Singh, Kelly A. Aschbrenner, Madeline E. Davies, Leslie Pelton-Cairns, and Gina R. Kruse. 2023. Challenges and recommendations for measuring time devoted to implementation and intervention activities in health equity-focused, resource-constrained settings: a qualitative analysis. *Implementation Science Communications* 4 (2023), 108. doi:10.1186/s43058-023-00491-7
[33] Sean Mann. 2024. Negative spillover due to constraints on care delivery: a potential source of bias in pragmatic clinical trials. *Trials* 25, 1 (2024), 833.
[34] Miruna Oprescu, Vasilis Syrgkanis, and Zhiwei Steven Wu. 2019. Orthogonal Random Forest for Causal Inference. In *Proceedings of the 36th International Conference on Machine Learning*, Vol. 97. PMLR, 4932–4941.
[35] Ashruta Patel. 2022. Improving chronic disease self-management in high-risk patient populations. *Translational Medicine Communications* 7 (2022), 4. doi:10.1186/s41231-022-00112-w
[36] Judea Pearl, Madelyn Glymour, and Nicholas P. Jewell. 2016. *Causal Inference in Statistics: A Primer.* Wiley, Hoboken, NJ.
[37] Iain Robertson-Steel. 2006. Evolution of triage systems. *Emergency Medicine Journal* 23, 2 (2006), 154–155.
[38] Juan D. Rodríguez, João Gama, and José R. Quevedo. 2021. Uplift modeling for preventing student dropout in higher education. *IEEE Transactions on Learning Technologies* 14, 3 (2021), 356–366. doi:10.1109/TLT.2021.3051387
[39] Donald B. Rubin. 2005. Causal inference using potential outcomes: Design, modeling, decisions. *J. Amer. Statist. Assoc.* 100, 469 (2005), 322–331.
[40] Vibhhu Sharma and Bryan Wilder. 2024. Comparing Targeting Strategies for Maximizing Social Welfare with Limited Resources. arXiv preprint arXiv:2411.07414.
[41] Mahmoud Shoush and Marlon Dumas. 2025. Prescriptive Process Monitoring Under Resource Constraints: A Reinforcement Learning Approach. *KI – Künstliche Intelligenz* 39 (2025), 119–140. doi:10.1007/s13218-024-00881-6
[42] Kebaso Sianyo, Jacob Kirera, Beryl Ondiso, and Javan Otieno. 2024. Leveraging AI To Bridge The Neurosurgeon Shortage In Africa. *East African Journal of Neurological Sciences* 3, Supp 1 (2024). https://theeajns.org/index.php/eajns/article/view/285
[43] Mohammed Usman and Ameena Banu. 2019. A study on impact of financial stress on students' academics. *Journal of Business and Economics Policy* 6 (2019), 58–64.
[44] Yixin Wang and José R. Zubizarreta. 2020. Minimal dispersion approximately balancing weights: asymptotic properties and practical considerations. *Biometrika* 107, 1 (2020), 93–105. doi:10.1093/biomet/asz050
[45] Travis T. York, Charles W. Gibson, and Susan Rankin. 2015. Defining and Measuring Academic Success. *Practical Assessment, Research, and Evaluation* 20, 5 (2015). doi:10.7275/hz5x-tx03

A Additional Robustness and Fairness-Constrained Allocation

This appendix reports two additional analyses that address robustness concerns about the Prospective IMPACT score and the role of fairness in the allocation rule. The first analysis varies the composite-score weights. The second analysis adds an optional fairness-constrained selection layer on top of the Prospective IMPACT score.

A.1 Composite-score weight sensitivity

The main Prospective IMPACT score is

$$A_i = z(G_i) + 0.25z(S_i^{med}) - 0.25z(C_q(X_i)). \tag{6}$$

The gain term G_i is the dominant component. The mediator-response and consistency terms are intended as stabilizers rather than tuned hyperparameters. To test whether the main findings depend on the specific value 0.25, we evaluate the generalized score

$$A_i(w_m, w_c) = z(G_i) + w_m z(S_i^{med}) - w_c z(C_q(X_i)), \tag{7}$$

where $w_m, w_c \in \{0, 0.10, 0.25, 0.50, 1.00\}$. This produces 25 score specifications at the main threshold $\theta = 0.70$.

Table 7 shows that the main conclusion is not driven by a narrow choice of auxiliary weights. At the 10% budget, all 25 score specifications outperform direct CATE in improved-count comparisons; the median specification identifies 12 threshold-crossing students and the best specification identifies 14, compared with 4 under direct CATE. At the 30%

Table 7. Composite-score weight sensitivity for Prospective IMPACT at the main threshold $\theta = 0.70$. The score is $A_i(w_m, w_c) = z(G_i) + w_m z(S_i^{med}) - w_c z(C_q(X_i))$, with $w_m, w_c \in \{0, 0.10, 0.25, 0.50, 1.00\}$.

Budget	Specs	Improved min	Improved med.	Improved max	Direct CATE	Beat	Loss
10%	25	7	12	14	4	100.0%	0.0%
30%	25	13	16	16	12	100.0%	0.0%

budget, all 25 specifications also outperform direct CATE; the median and maximum specifications identify 16 threshold-crossing students, compared with 12 under direct CATE. The canonical $(0.25, 0.25)$ choice is therefore interpreted as a moderate stabilizing specification, not as an ex post tuning choice.

A.2 Fairness-constrained Prospective IMPACT

The main paper treats protected attributes as auditing variables. This is the conservative specification because protected-attribute-aware allocation may face legal and institutional constraints. As an extension, we also evaluate a fairness-constrained allocation layer. Let $z_i \in \{0, 1\}$ indicate whether at-risk student i is selected and let $n_b = \lceil b|\mathcal{I}_{risk}|\rceil$. For a candidate selected set $\mathcal{S}_b = \{i : z_i = 1\}$, define the within-race selection-rate gap as

$$\Delta(\mathcal{S}_b) = \max_g \frac{|\mathcal{S}_b \cap g|}{|\mathcal{I}_{risk} \cap g|} - \min_g \frac{|\mathcal{S}_b \cap g|}{|\mathcal{I}_{risk} \cap g|}. \tag{8}$$

The constrained extension solves the score-maximization problem

$$\max_{z_i \in \{0,1\}} \sum_{i \in \mathcal{I}_{risk}} z_i A_i \quad \text{s.t.} \quad \sum_{i \in \mathcal{I}_{risk}} z_i = n_b, \quad \Delta(\mathcal{S}_b) \le \delta. \tag{9}$$

The parameter δ controls the maximum allowed racial selection-rate gap. We evaluate $\delta \in \{0.025, 0.05, 0.075, 0.10, 0.15, 0.20\}$ and report the most policy-relevant cases below.

Table 8 shows that explicit fairness constraints can sharply reduce allocation gaps with little or no loss in predicted utility. At a 10% budget, imposing $\delta = 0.05$ preserves the same 13 improved students as unconstrained Prospective IMPACT while reducing the selection-rate gap from 0.050 to 0.040. A stricter constraint, $\delta = 0.025$, reduces the gap to 0.019 while selecting 12 improved students. At a 30% budget, both reported constrained variants keep 16 improved students, equal to the unconstrained rule, while reducing the selection-rate gap from 0.169 to 0.020 or 0.040.

These results show that the framework can support both auditing and explicit constrained allocation. The main specification remains audit-only because it avoids using protected attributes in the allocation rule. The constrained variant is best interpreted as an optional policy layer for settings where explicit group constraints are legally and institutionally appropriate.

A.3 Practical Interpretation

The ELS results suggest a practical workflow for institutional decision support. A school, college, or public agency can define an eligible at-risk population, specify a budget and success threshold, and compare candidate allocation rules on the same population. Prospective IMPACT is useful because it respects the timing of real allocation decisions. It does not use the candidate student's first-year GPA or earned credits at scoring time, but it still asks whether historically plausible mediator pathways are predicted to change final success.

This transparency is important for accountability. The framework decomposes the basis for allocation into baseline risk, mediator responsiveness, predicted final-outcome gain, and subgroup allocation patterns. A financial-aid office

Table 8. Fairness-constrained Prospective IMPACT. The constrained variant maximizes the Prospective IMPACT score subject to an upper bound δ on the within-race selection-rate gap.

Budget	Method	Improved	Avg. gain	Selection gap
10%	Prospective IMPACT	13	0.279	0.050
10%	Direct CATE	4	0.072	0.201
10%	Constrained, $\delta = 0.025$	12	0.277	0.019
10%	Constrained, $\delta = 0.050$	13	0.278	0.040
30%	Prospective IMPACT	16	0.189	0.169
30%	Direct CATE	12	0.061	0.274
30%	Constrained, $\delta = 0.025$	16	0.188	0.020
30%	Constrained, $\delta = 0.050$	16	0.189	0.040

may ask whether limited grants are being directed toward students whose academic progress is likely to change. An equity review committee may ask whether the selected set disproportionately excludes a racial group or whether predicted benefits accrue unevenly. IMPACT is designed to make these questions part of the allocation workflow rather than after-the-fact explanations.

The framework also distinguishes allocative evidence from allocative authority. The empirical exercise can show that, under a specified objective, some students are predicted to be more responsive than others. It should not be read as saying that only those students deserve institutional support. Public agencies and universities may legitimately adopt additional criteria, including need, legal constraints, minimum service guarantees, institutional mission, or commitments to historically underserved populations. IMPACT supports deliberation about tradeoffs rather than hiding those tradeoffs inside a single score.

A.4 Uncertainty and Robustness

The sensitivity analyses clarify the role of researcher- and institution-chosen parameters. The success threshold θ changes the definition of the at-risk set and the definition of improvement. The matching parameter k changes how local the historical comparison is. The auxiliary score weights change how much mediator response and local consistency stabilize the predicted final-outcome gain. The fairness penalty λ changes the weight assigned to racial allocation imbalance. A credible allocation framework should not depend on a single convenient value of any of these parameters. Across the tested ranges, Prospective IMPACT remains the strongest or a highly competitive benefit-targeting allocation and maintains favorable group-level allocation diagnostics relative to the main baselines.

The goal of these robustness checks is not to claim exact numerical precision for each selected student. The ELS setting is observational, and all allocations are estimated rather than directly observed causal truth. The checks show that the comparative ranking is not an artifact of one threshold, one matching-neighbor choice, one auxiliary score weighting, or one fairness-penalty value.

A.5 Scope Beyond Financial Aid

Although the empirical application is financial aid, the structure of the framework is broader. Many public-sector interventions operate through intermediate outcomes that are observed before the final outcome. In workforce programs, a training subsidy may affect course completion, certification, or job-search intensity before employment is observed. In public health, an outreach intervention may affect screening uptake, medication adherence, or follow-up attendance before longer-run health outcomes emerge. In social services, housing assistance or case-management programs may affect intermediate stability measures before final outcomes such as employment, safety, or school attendance are realized.

The main methodological requirement is not that the same learners or covariates be used in every domain. Rather, the analyst must be able to distinguish baseline covariates from post-treatment mediators, estimate or reconstruct mediator responses, and evaluate whether predicted mediator changes are consequential for the final outcome. The risk model, mediator CATE estimators, matching layer, outcome model, and fairness metrics can be replaced or extended without changing the overall decision logic. The allocation and auditing layer remains the same: compare candidate policies under the same budget, estimate who is predicted to benefit, and audit whether allocation and benefit patterns differ across groups.

B Ethical Considerations and Limitations

IMPACT is intended to support, not replace, human decision-making. Allocation rules for financial aid and educational support can affect access to opportunity, so any deployment would require institutional oversight, transparency, and opportunities for review or appeal. The ELS analysis uses de-identified secondary data; it does not recruit participants, contact students, or make real financial-aid decisions. Sensitive demographic attributes are used only for evaluation and auditing in the main specification. Race is not used to claim that the allocation is fair; rather, it is used to examine whether selection and predicted benefit patterns differ across groups. Any deployment or extension to restricted institutional data would require compliance with applicable data-use agreements, legal constraints, and institutional review requirements.

Several limitations remain. First, the causal interpretation depends on observed covariates being sufficient to address confounding and on assumptions about mediator-outcome relationships. These assumptions are strong in the ELS financial-aid setting because grant receipt may depend on unobserved factors such as application behavior, parental guidance, institutional discretion, unmet need, or local aid rules. Propensity overlap and covariate-balance diagnostics reduce concerns about observed imbalance, but they cannot rule out unobserved confounding. The ELS results should therefore be interpreted as estimated allocation benefits under observed-covariate adjustment, not as experimental causal effects.

Second, Prospective IMPACT avoids using the candidate student's realized post-treatment mediators at scoring time, but it still depends on the quality of baseline matching and the availability of comparable historical treated and untreated students. If matched treated and untreated students differ on unobserved determinants of mediator response, the reconstructed proxy mediator profiles may be biased. Third, the outcome-aware score depends on model quality. We report AUC, Brier scores, overlap, threshold sensitivity, matching-neighbor sensitivity, score-weight sensitivity, and fairness-penalty sensitivity, but these diagnostics do not prove causal identification or prospective calibration.

Finally, fairness has multiple definitions. We report selection-rate gaps, benefit shares, improvement rates, mean predicted gains within groups, and fairness-adjusted utility, but these metrics do not exhaust the ethical dimensions of educational resource allocation. The fairness-constrained extension illustrates one way to impose explicit equity constraints, but protected-attribute-aware allocation may face legal and institutional restrictions and should be evaluated in context. Real deployment would also require assessment of administrative feasibility, appeal procedures, institutional capacity, temporal drift, prospective calibration, and behavioral responses to allocation rules.